

SE3082

AWS EC2 Lab - Screenshots and Calculations

Lab 06

Exercise 1: Pragma omp (openmp) barrier code

IT24610328

On instance with 2 vCPUs (t3.micro)

Nano - creating the pragma omp barrier code

```
GNU nano 6.2                                ex1_barrier.c
#include <omp.h>
#include <stdio.h>

int main() {

#pragma omp parallel
{
    int th = omp_get_thread_num();
    printf("thread # %d\n", th);

    #pragma omp barrier

    printf("Outside Barrier # %d\n", omp_get_thread_num());
    printf("Hi again from thread # %d\n", omp_get_thread_num());
}

    return 0;
}
```

Outputs from running the pragma omp barrier code.

```
[ec2-user@ip-172-31-17-201 ~]$ nproc --all
2
[ec2-user@ip-172-31-17-201 ~]$ gcc -fopenmp ex1_barrier.c -o ex1_barrier.o
[ec2-user@ip-172-31-17-201 ~]$ ./ex1_barrier.o
thread # 1
thread # 0
Outside Barrier # 1
Hi again from thread # 1
Outside Barrier # 0
Hi again from thread # 0
[ec2-user@ip-172-31-17-201 ~]$ ./ex1_barrier.o
thread # 1
thread # 0
Outside Barrier # 1
Hi again from thread # 1
Outside Barrier # 0
Hi again from thread # 0
[ec2-user@ip-172-31-17-201 ~]$ ./ex1_barrier.o
thread # 0
thread # 1
Outside Barrier # 0
Hi again from thread # 0
Outside Barrier # 1
Hi again from thread # 1
```

How the `pragma omp barrier` code works in the program.

`#pragma omp barrier` makes every thread stop at that line and wait until all other threads in the parallel region have also reached it. No thread continues past the barrier until every thread has arrived. That's why all the "thread # x" lines print first (each thread's individual work before the barrier), then only once every thread has hit the barrier do the "Outside Barrier" and "Hi again" lines start appearing, in whatever order threads happen to resume.